

Software Requirements Specification

For

TOURHUB

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Introduction

The Tour Booking System is a software application that simplifies and streamlines the management of travel and tourism services. This system offers a central platform where agencies, consultants, and customers can interact effectively for tour planning, booking, and management. It reduces the manual work typically needed in traditional booking methods and improves communication among users. Through this platform, customers can easily search for and book tour packages. Consultants can offer travel-related services like hotels, flights, visa applications, and insurance. Agencies can manage the entire booking process and keep track of system activities.

In today's tourism industry, online booking systems are crucial for boosting customer satisfaction and improving operational efficiency. Clearly defining the system requirements is vital for developing and implementing the Tour Booking System successfully. A well-prepared Software Requirements Specification (SRS) document helps developers, designers, and stakeholders understand the system's functions, features, limitations, and expected performance. It also serves as a reference throughout the software development life cycle to ensure the final system meets user needs and business goals.

Purpose

The purpose of this Software Requirements Specification (SRS) document is to describe the Tour Booking System and its requirements in detail. This document outlines the functional and non-functional requirements needed for the system's development and implementation. It provides guidelines for developers, testers, designers, and stakeholders so they can understand how the software should behave and what features it should offer.

The SRS document aims to capture all major system requirements, including user interactions, booking management, payment processing, notifications, refund handling, and feedback management. It ensures that all stakeholders share a common understanding of the system's requirements before development starts.

Scope

The Tour Booking System focuses on creating an online platform for managing tour packages, booking services, customer interactions, consultant services, and administration. It aims to handle the entire travel booking process in a secure and efficient way.

The system offers key features like user login and profile management, searching and filtering tour packages, viewing package details and pricing, booking tours and guides, processing payments, managing cancellations and refunds, sending notifications, displaying advertisements, collecting feedback and ratings, managing consultants, providing visa application services, offering travel insurance, and confirming bookings. It also enables agencies to track activities and manage consultants and customer feedback effectively.

The system is limited to online tour booking and travel service management. It does not include offline booking, advanced AI-based recommendations, or third-party logistics management in its current scope.

It serves three types of users:

- Agency (Administrator/Stakeholder)
- Consultant (Service Provider)
- Customer (User/Traveler)

Each user has different access rights and features based on their role in the system.

Objectives

The main goal of the Tour Booking System is to create an efficient, secure, and user-friendly platform for handling tourism and travel booking services. The system focuses on making the booking process easier and improving coordination among agencies, consultants, and customers using digital tools.

The objectives of the system are:

- Reduce manual work involved in traditional tour booking methods.
- Provide customers with easy access to tour packages and booking services.
- Improve communication between agencies, consultants, and customers.
- Offer secure online payment, cancellation, and refund services.
- Help agencies manage tour packages, consultants, advertisements, and feedback effectively.
- Increase customer satisfaction by providing reliable and convenient travel services.
- Boost operational efficiency and accuracy in managing travel and tourism activities.

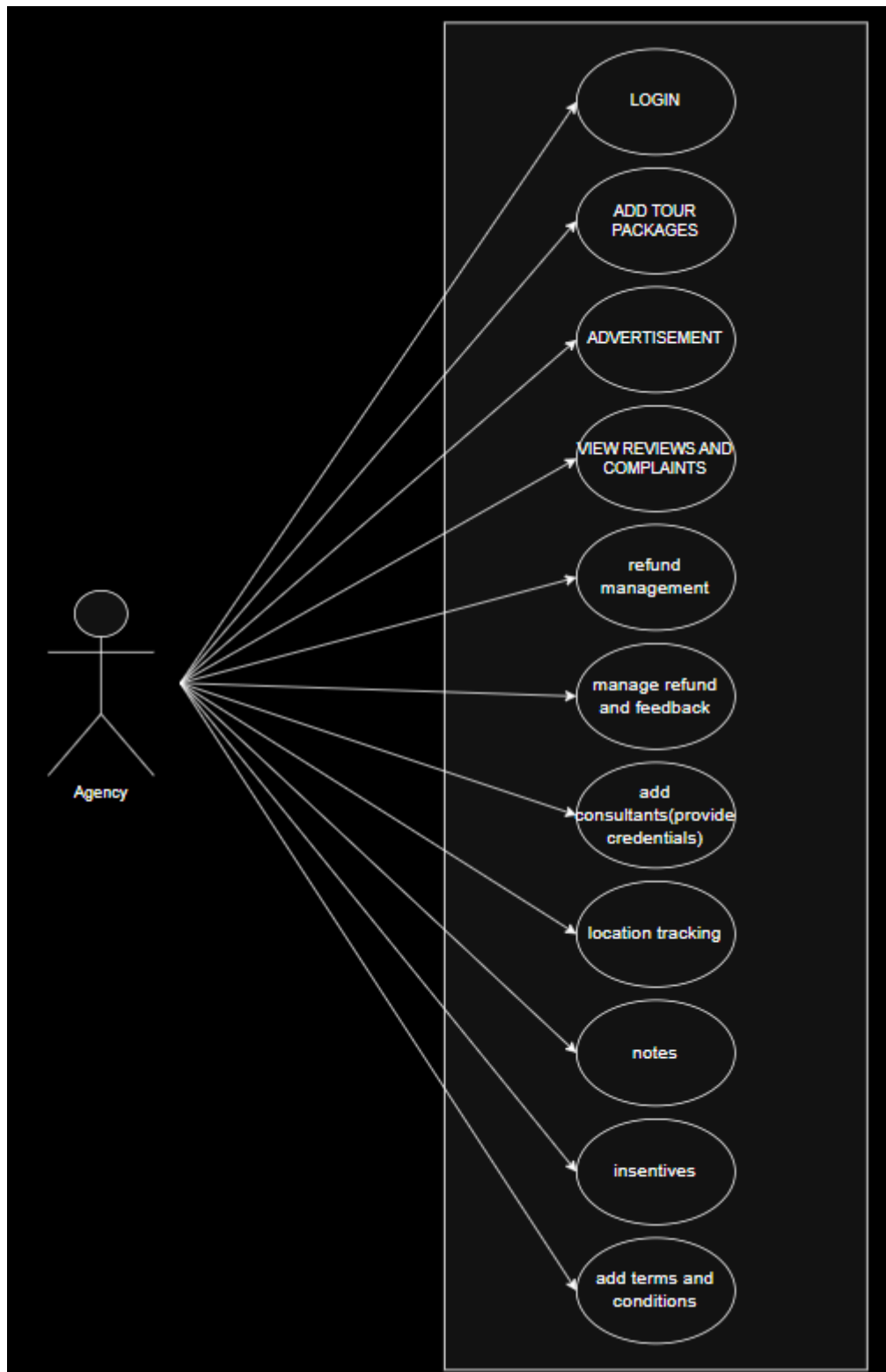
Functional Requirements

Primary actors of the system

- Agency**
- Consultant**
- Customer**

Use Case Diagrams

Agency



Consultant

Login

Feedback

Monitor Reviews

Reserve Tickets

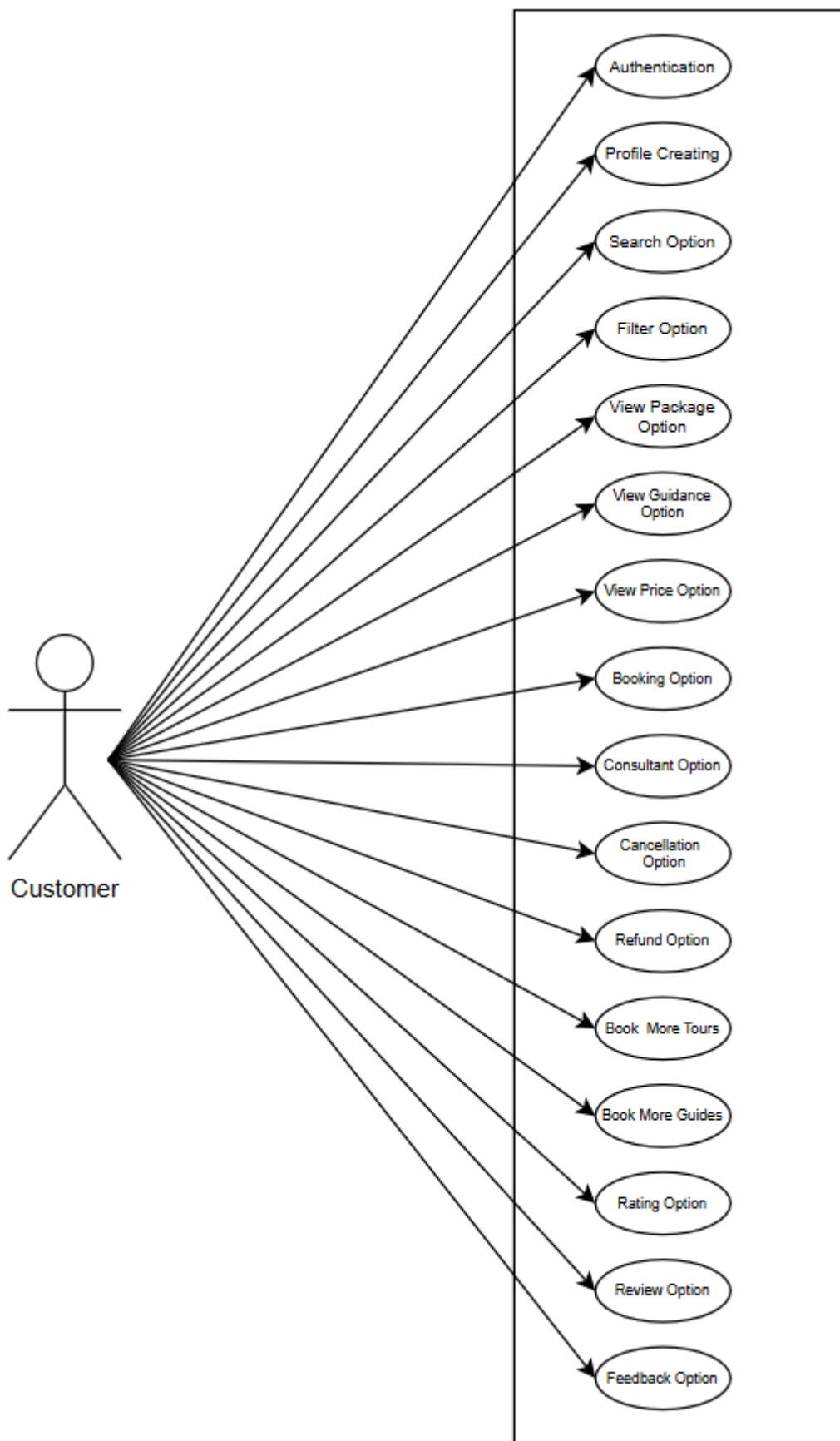
Visa Application

Travel Insurance

Available slots

Tour Packages

Notifications



User Stories

Agency

As a (user role)	I want (goal)	So that (benefit)
AGENCY	I want to login	So that I can manage the entire process
AGENCY	I want to add tour packages	So that I can provide details of tour packages
AGENCY	I want to give advertisement	So that I can promote this tour agency
AGENCY	I want to view reviews and complaints	So that I can improve the quality of service
AGENCY	I want to manage refund activities	So that I can satisfy the customer and improve customer trust
AGENCY	I want to manage feedback	So that I can delete any bad feedbacks
AGENCY	I want to add consultants(provide credentials)	So that they can login and work under the agency and manage consultant activities
AGENCY	I want to track the location	So that I can know the current location of customers and ensure safe and efficient travel
AGENCY	I want to add notes	So that I can provide additional information to the customer
AGENCY	I want to give incentive to the consultants	So that I can encourage consultants to work efficiently
AGENCY	I want to add terms and conditions	so that they can know the rules for using this web application

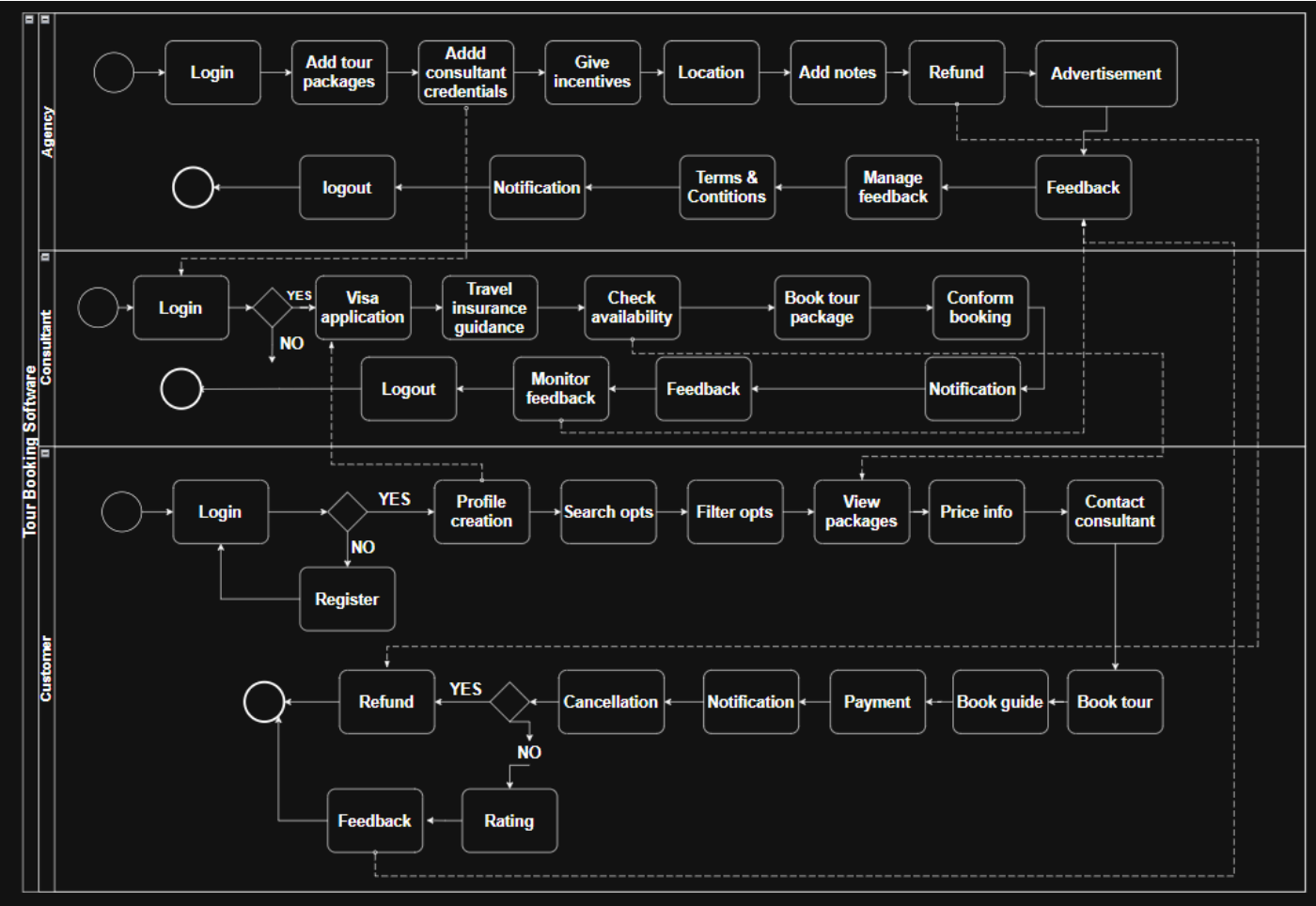
Consultant

As a	I need	So that
Consultant	To allow users to register and log in to the application	Customers can safely access their accounts and manage bookings easily
Consultant	To create a feedback form for customers	Customers can share their opinions and help improve the service
Consultant	To monitor reviews and complaints	Customer problems can be solved quickly and service quality can improve
Consultant	To provide an option for reserving tickets	Customers can book tickets easily and save time
Consultant	To assist customers with the visa application process	Customers can get support and guidance for visa approval
Consultant	To provide travel insurance services	Customers can travel with financial protection and peace of mind
Consultant	To display tour vacancies and availability	Customers can quickly check available slots before booking
Consultant	To add and manage tour packages	Customers can choose tour packages that fit their needs and budget
Consultant	To send text and email notifications	Customers can receive booking updates and reminders on time

Customer

<i>As a (user role)</i>	<i>I want (goal)</i>	<i>so that (benefit)</i>
Customer	<i>Register into the application by entering details.</i>	<i>Make my data available to agency and use the system capabilities reserved for registered users.</i>
Customer	<i>I want to create my profile with my interests</i>	<i>So that the agency can easily provide tour details to me that can match my interests.</i>
Customer	<i>I want a search option</i>	<i>So that I can quickly discover tours worldwide without browsing endlessly.</i>
Customer	<i>I want a view packages feature</i>	<i>So that I get the best value and avoid hidden surprises.</i>
Customer	<i>I want view guidance availability</i>	<i>So that I can choose tours with expert guides, enhancing my experience with local insights and cultural knowledge.</i>
Customer	<i>I want to view price information</i>	<i>So that I can make budget effectively, and avoid unexpected costs.</i>
Customer	<i>I want a booking option</i>	<i>So that I can secure my spot instantly and avoid missing out on popular tours.</i>
Customer	<i>I want consultant contact option</i>	<i>So that I can get expert to clarify doubts and feel confident before committing to a tour.</i>
Customer	<i>I want a cancellation option</i>	<i>So that I can cancel my tour if necessary.</i>
Customer	<i>I want a refund option</i>	<i>So that I don't lose money when cancelling, making the service trustworthy and risk-free.</i>
Customer	<i>I want a book more tours /guides option</i>	<i>So that I can easily extend my travel experience, bundle multiple tours and customise my tour with additional expert guides if necessary.</i>
Customer	<i>I want a rating, review & feedback option</i>	<i>So that I can share my opinion, document my experience in details to contribute to improving tour quality and help the future travellers make better choices.</i>

Business Process Diagram



Non-Functional Requirements

The non-functional requirements define the quality attributes and operational standards of the Tour Booking System. These requirements ensure that the system performs efficiently, securely, reliably, and provides a satisfactory user experience for agencies, consultants, and customers.

1. Performance Requirements

- The system should provide fast response times for user requests such as searching tour packages, booking tours, and processing payments.
- The system should support multiple users accessing the platform simultaneously without significant delay.
- Page loading time should be minimized to improve user experience.
- The booking confirmation and notification processes should be completed within a few seconds after successful transactions.
- The database should efficiently manage large volumes of customer, consultant, and booking information.

2. Reliability Requirements

- The system should operate continuously with minimal downtime.
- Data entered by users should be stored accurately without loss or corruption.
- The system should provide backup and recovery mechanisms to protect important data.
- In case of system failure, recovery procedures should restore the system quickly and securely.
- The booking and payment processes should maintain consistency and avoid duplicate transactions.

3. Security Requirements

- The system should provide secure login authentication for all users.
- User passwords should be encrypted and securely stored in the database.

- Only authorized users should be allowed to access specific modules and functionalities.
- Online payment transactions should be protected using secure payment gateways and encryption techniques.
- The system should prevent unauthorized access, data manipulation, and cyber threats.
- User personal information and payment details must remain confidential.

4. Usability Requirements

- The system should provide a simple, user-friendly, and easy-to-navigate interface.
- Users should be able to access features without requiring technical knowledge.
- The system should provide clear instructions, notifications, and error messages.
- Search and filter options should help customers quickly find suitable tour packages.
- The interface should be visually organized and responsive across different devices.

5. Availability Requirements

- The system should be available to users 24/7 for booking and management activities.
- Scheduled maintenance should be performed during low-traffic periods to minimize disruption.
- The system should ensure high availability during peak booking seasons.

6. Scalability Requirements

- The system should support future expansion of users, tour packages, and services.
- Additional features such as advanced analytics or AI-based recommendations should be easily integrated in future versions.

- The database and server infrastructure should handle increasing workloads efficiently.

7. Maintainability Requirements

- The system should be designed using modular architecture for easier maintenance and updates.
- Errors and bugs should be easy to identify and correct.
- System documentation should be maintained properly for future development and troubleshooting.
- New functionalities should be added without affecting existing operations.

8. Compatibility Requirements

- The system should be compatible with major web browsers such as Chrome, Firefox, Edge, and Safari.
- The platform should support desktop computers, laptops, tablets, and mobile devices.
- The system should integrate properly with payment gateways and notification services.

9. Data Management Requirements

- The system should securely store customer, consultant, booking, and payment records.
- Data should be organized and easily retrievable when required.
- The system should maintain data integrity and avoid duplicate records.
- Proper database management techniques should be implemented for efficient performance.

10. Legal and Compliance Requirements

- The system should follow applicable data protection and privacy regulations.
- Terms and conditions must be clearly displayed for users before booking confirmation.

- The system should maintain transaction records for auditing and administrative purposes.
- Refund and cancellation policies should be handled according to agency rules and regulations.

Technical Requirements

The technical requirements define the hardware, software, platform, and technology specifications needed for the successful development, deployment, and operation of the Tour Booking System. These requirements ensure that the system functions efficiently, securely, and reliably across different environments and devices.

1. Hardware Requirements

Server Requirements

- Processor: Intel Core i5 or higher
- RAM: Minimum 8 GB
- Storage: Minimum 500 GB Hard Disk or SSD
- Network: Stable high-speed internet connection
- Backup storage for data recovery and protection

Client Requirements

- Processor: Dual-core processor or higher
- RAM: Minimum 4 GB
- Storage: Minimum 100 GB free space
- Display: Minimum screen resolution of 1024 × 768
- Internet connection for accessing the online system

2. Software Requirements

Operating System

- Windows 10/11
- Linux
- macOS

Web Browsers

- Google Chrome

- Mozilla Firefox
- Microsoft Edge
- Safari

Database Management System

- PostgreSQL for storing user, booking, payment, and travel data

Development Tools

- Visual Studio Code
- Git for version control

3. Programming Languages

The Tour Booking System may be developed using the following programming languages:

Front-End Technologies

- HTML5
- CSS3
- JavaScript

Back-End Technologies

- Python

Database Query Language

- SQL

4. Frameworks and Libraries

Front-End Frameworks

- Bootstrap
- React

Back-End Frameworks

- Fast API(for Python)

Additional Libraries and APIs

- Payment Gateway APIs for online payment processing
- Notification APIs for email and SMS alerts
- Google Maps API for location services

5. Platform Requirements

- The system should operate as a web-based application accessible through internet browsers.
- The application should support both desktop and mobile platforms.
- The system should be deployable on cloud or local servers.

6. Compatibility Requirements

- The system should be compatible with multiple operating systems and browsers.
- The user interface should be responsive across desktops, tablets, and smartphones.
- The system should support integration with third-party services such as payment gateways and travel-related APIs.

7. Security Requirements

- Passwords should be securely encrypted in the database.
- Secure authentication and authorization mechanisms should be implemented.

8. Network Requirements

- Continuous internet connectivity is required for accessing the system.
- The server should support secure online transactions and real-time notifications.
- Adequate bandwidth should be available to support multiple concurrent users.

9. Performance Requirements

- The system should support multiple users simultaneously without performance degradation.
- Search, booking, and payment operations should respond quickly.
- The database should efficiently manage large amounts of data.

10. Maintenance and Support Requirements

- The system should support regular updates and bug fixes.
- Proper technical documentation should be maintained.
- The software architecture should allow future enhancements and scalability.

Conclusion

The Software Requirements Specification (SRS) document for the Tour Booking System provides a detailed description of the system requirements, functionalities, objectives, technical specifications, and non-functional requirements needed for the successful development and implementation of the application. The document clearly defines the roles and responsibilities of the three major users of the system: the Agency, the Consultant, and the Customer. It also explains the various services and features provided by the system, including tour package management, booking services, payment processing, notifications, feedback management, visa applications, travel insurance, cancellations, and refunds.

The SRS document serves as an important guideline for developers, designers, testers, and stakeholders throughout the software development life cycle. By clearly identifying the functional and technical requirements, the document helps reduce misunderstandings, improve communication among stakeholders, and ensure that the developed system meets user expectations and business objectives.

The Tour Booking System aims to provide a centralized, secure, reliable, and user-friendly platform that simplifies travel and tourism management activities. The system is expected to improve operational efficiency, reduce manual work, enhance customer satisfaction, and provide better coordination between agencies, consultants, and customers.

For successful implementation, it is recommended that the system be developed using modern technologies, secure payment integration, responsive design principles, and proper database management techniques. Regular maintenance, security updates, and future scalability considerations should also be included to ensure long-term performance and reliability of the system.

Overall, the Tour Booking System will contribute significantly to modernizing travel booking operations and improving the overall experience for all users involved in the tourism industry.